

Memory-Aware Software Engineering Agents: Architectures, Mechanisms, and Open Gaps

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Abstract. AI-assisted software engineering has progressed from code completion to agentic systems operating across the full development lifecycle. Yet these agents remain fundamentally stateless, i.e. each task begins without memory of prior sessions, resolved bugs, or accumulated project understanding. In particular, they lack episodic memory, i.e. structured records of past sessions, debugging trajectories, and decision rationale, as well as temporal versioning, i.e. the ability to track when a piece of knowledge was valid, when it changed, and when it became stale. We present a survey that synthesizes evidence from a feature-level analysis of ten production software engineering harnesses with a specific focus on the current state of the art in memory architecture for software engineering agents research. Our analysis reveals an episodic-temporal deficit, i.e. no production harness ships episodic memory or temporal versioning natively. Interestingly, a rapidly growing ecosystem of add-ons based on the Model Context Protocol (MCP) has emerged to fill the deficit. Yet, even these rarely implement temporal reasoning. Similarly, within the research community, while addressing episodic memory only a small fraction consider temporal mechanisms. At the same time, benchmarks confirm a factual-to-cognitive gap across production memory architectures. We argue that persistent, temporally grounded memory is the prerequisite for software engineering agents that learn alongside the teams they serve.

Keywords: Software Engineering Agents · Agentic Memory · Context Engineering · Temporal Reasoning · Knowledge Graphs

1 Introduction

Agentic AI systems now operate across the full software engineering lifecycle [1,2,3]. Whether resolving complex bugs through multi-file analysis, discovering architectural patterns, or coordinating code reviews across a team, these agents accumulate significant task-specific knowledge during each session. Yet that knowledge vanishes when the session ends, i.e. the next invocation begins from scratch, repeating investigations, rediscovering constraints, and re-learning project conventions. Recent production systems have begun to address this gap, e.g. Claude Code persists automatically extracted memories across sessions and Copilot injects repository-level custom instructions as reusable project context [7], but these mechanisms remain limited in scope and depth.

The field has responded rapidly [5,6], with every major AI provider now shipping some form of persistent memory. Yet no production harness implements episodic memory or temporal versioning natively. A thriving ecosystem of community-built MCP add-ons [38] has emerged, evidence that practitioners feel the deficit acutely, but fundamental properties such as team-shared memory or decision provenance are only rarely addressed.

In this paper, we survey and classify memory architectures for software engineering agents. Specifically, we i) conduct a feature-level analysis of ten production software engineering harnesses and ten representative MCP-based community add-ons across 20 dimensions; ii) classify memory architectures from the recent research literature; and iii) identify open gaps at the intersection of memory research and software engineering practice. Our goal is to provide a structured overview of the current state of the art and to highlight where the most impactful research and engineering efforts are needed.

2 Memory in Production Software Engineering Harnesses

The CoALA taxonomy [4] maps human memory types onto agent components, i.e. *working*, *semantic*, i.e. conventions, facts, *episodic*, such as past debugging experiences, and *procedural*, such as workflows or tool usage. Thereby, working memory is well-served by modern context windows and semantic memory is partially addressed by context files [7]. However, episodic and procedural memory are almost entirely absent. A finer-grained functional taxonomy [5] distinguishes *factual*, i.e. what an agent knows, from *experiential*, i.e. what the agent has learned from doing, and *working* memory, i.e. active context management, with all three being critical for software engineering agents operating over long project horizons.

We selected ten production software engineering agent harnesses (see Table 1) based on three criteria: i) general availability as of June 2026, ii) active development with at least one major update in the preceding six months, and iii) widespread adoption. From over 40 MCP-based memory servers, we selected ten representative examples spanning different CoALA memory types. Each system was evaluated through official documentation and publications, hands-on testing where accessible, and community reports. Capabilities were rated as full support, partial support, or absent; ambiguous cases were rated conservatively.

We evaluate along 20 dimensions grouped into six categories. The first three, *CoALA memory types*, *memory operations*, and *temporal mechanisms*, are derived from the CoALA [4] and Hu et al. [5] taxonomies. The remaining three, *SE-specific capabilities*, *collaboration & governance*, and *integration*, extend the evaluation to practical concerns specific to software engineering workflows. Table 1 summarizes the comparison. All ten harnesses support human-written instruction files, representing declarative semantic memory, with widespread adoption [7], yet little resolution improvement [8]. Context engineering [45] formalizes the approach, but all implementations remain manually authored without temporal metadata. Five harnesses generate memories automatically (Claude Code, Codex CLI,

Copilot, Windsurf, Gemini CLI), but none distinguishes factual from experiential knowledge. Context compaction triggers vary with respect to window utilization, however implementations are lossy, i.e. technical details, intermediate reasoning steps, and failed-approach information are discarded. However, the quality of compaction determines whether memory helps or hinders [21].

No production system supports temporal versioning or episodic memory. Notably, those memory capabilities most critical for software engineering agents that learn over time are those that no tool implements. This gap has driven a community response, as described next.

	Memory Types			Memory Operations				Temporal			Software Engineering					Collaboration & Governance			Integration	
	Semantic	Episodic	Procedural	Auto-Generated	Compaction	Fact./Experiential	Staleness	Temporal Metadata	Temporal Versioning	Bi-Temporal	Code-Structural	Cross-Session	Provenance	Failed Replay	Git-Aware	Team/Collab.	Access Control	Memory Provenance	MCP Support	Local Deploy
Production Agents																				
🔗 Claude Code	✓	✗	✗	✓	✓	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✓	✓
🔗 Gemini CLI	✓	🟡	🟡	✓	✓	✗	✗	🟡	✗	✗	🟡	🟡	✗	✗	✗	✗	✗	✗	✓	✓
🔗 Codex CLI	✓	✗	✗	✓	✓	✗	🟡	🟡	✗	✗	✗	🟡	✗	✗	✗	✗	✗	✗	✓	✓
🔗 Copilot	✓	✗	✗	✓	✓	✗	🟡	🟡	✗	✗	🟡	✗	✗	✗	✗	✗	✗	✗	✓	✗
🔗 Cursor	✓	✗	✗	✗	✓	✗	✗	✗	✗	✗	🟡	✗	✗	✗	✗	✗	✗	✗	✓	✗
🔗 OpenCode	✓	✗	✗	✗	✓	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✓	✓
🔗 Windsurf	✓	✗	✗	✓	🟡	✗	✗	✗	✗	✗	🟡	✗	✗	✗	✗	✗	✗	✗	✓	✗
🔗 Devin	🟡	✗	✗	🟡	🟡	✗	✗	✗	✗	✗	🟡	✗	✗	✗	✗	🟡	✗	✗	✓	✗
🔗 Aider	🟡	✗	✗	✗	✗	✗	✗	✗	✗	✗	🟡	✗	🟡	✗	✓	✗	✗	✗	🟡	✓
🔗 OpenHands	✓	✗	✗	✗	✓	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✓	✓
MCP Community Add-ons																				
🔗 claude-mem	✓	✓	✗	✓	✓	✗	✗	🟡	✗	✗	✗	✗	✓	✗	✗	✗	✗	✗	✓	✓
🔗 Mem0	✓	🟡	✗	✓	✓	✗	✗	🟡	✗	✗	✗	✓	✗	✗	✗	🟡	✗	✗	✓	✓
🔗 Graphiti / Zep	✓	✓	✗	✓	✓	✗	✓	✓	✓	✓	✗	✓	✓	🟡	✗	✗	✗	✓	✓	✓
🔗 Cognee	✓	✗	🟡	✓	✓	✗	✗	🟡	✗	✗	✓	✓	✗	✗	✗	✗	✗	🟡	✓	✓
🔗 AgentMemory	✓	✓	🟡	✓	✓	🟡	✗	🟡	✗	✗	Agent	✓	🟡	✗	✗	✗	✓	✓	✓	✓
🔗 Letta / MemGPT	✓	🟡	🟡	✗	✗	✗	🟡	🟡	✗	✗	✗	✓	✓	🟡	✗	✗	✗	🟡	✓	✓
🔗 Codebase-Memory	✓	✗	✗	✓	✗	✗	✗	✗	✗	✗	✓	✓	✗	✗	✗	✗	✗	✗	✓	✓
🔗 Engram-MCP	✗	✓	✓	✓	✓	✓	✗	✓	🟡	✗	✗	✓	✓	✓	✓	✗	✗	✗	✓	✓
🔗 Ourmem	✓	✗	✗	✓	🟡	✗	✗	🟡	✗	✗	✗	✓	✗	✗	✗	✓	✓	🟡	✓	🟡
🔗 Hindsight	✗	✓	✗	✓	🟡	✗	🟡	🟡	✗	✗	✗	✓	✓	🟡	✓	✗	✗	✗	✓	✓
✓ Supported 🟡 Partial ✗ Absent																				

✓ Supported ✗ Partial ✗ Absent

Table 1. Memory capability matrix across 10 production SE agent harnesses and 10 representative MCP community add-ons, evaluated along 20 dimensions grouped into six categories: CoALA memory types, memory operations, temporal mechanisms, SE-specific capabilities, collaboration & governance, and integration. Temporal versioning, bi-temporal validity, and governance remain near-universally absent.

2.1 Community Extensions via MCP

The gap between what production harnesses ship and what practitioners need has spawned a rapidly growing ecosystem of community-built memory extensions via MCP [38]. Our survey identifies over 40 MCP-based memory servers (see selection in table 1). The ecosystem spans every CoALA memory type, however with stark imbalances, i.e. semantic memory is near-universal, episodic recall is emerging, while temporal versioning and procedural memory remain nearly absent. Among the most adopted, *claude-mem* [40] hooks into Claude Code lifecycle events for session capture. *AgentMemory* [39] provides hybrid of BM25 keyword search, vector embeddings and knowledge graph retrieval and *Graphiti/Zep* [12] remains the only reviewed production system with true bi-temporal validity intervals. Software engineering specific servers include *Codebase-Memory* [36], a Tree-Sitter knowledge graph across a plethora of languages, *Engram-MCP* [41], for git-branch-aware session handoffs, or *Hindsight* [42], featuring failed attempt replay. Notably, nine of ten major coding agents now support MCP, making these add-ons agent-agnostic. Built-in memory sophistication varies widely. For example, Gemini CLI offers a four-tier hierarchy with automatic skills extraction, i.e. episodic-to-procedural conversion. Copilot performs just-in-time codebase verification of stored facts, while Cursor has removed its auto-memory feature, suggesting that naive auto-capture without proper architecture degrades quality.

3 Memory Architectures: From Extraction to Graphs

Our classification of the ICLR 2026 MemAgents workshop paper collection [6] reveals that retrieval-augmented memory is the dominant paradigm, underpinning over 60% of contributions. Within this paradigm, approaches differ primarily in how they organize, consolidate, and temporally ground stored memories. A second, smaller cluster of non-RAG approaches operates at the model or inference level, including KV-cache compression, attention-based, and embedding-based methods [5]. Cross-cutting learning strategies such as experience replay, reinforcement learning-based management, and hierarchical consolidation span both clusters. Three retrieval-augmented variants have reached production maturity and been formally benchmarked head-to-head on the LoCoMo-Plus cognitive memory evaluation [9], i.e. extraction-based, self-managing, and graph-based temporal. Table 2 summarizes their key properties and relates them to the evaluation dimensions of Table 1.

3.1 Extraction-Based Memory

Extraction-based architectures, such as Mem0 [10], extract atomic facts from conversations and store them as vector-indexed entries for similarity based querying. Extraction-based systems excel at *factual recall* and correspond to the *semantic memory* and *factual knowledge* columns in Table 1. For software engineering tasks requiring explicit knowledge retrieval, this paradigm is well-suited. However, extraction is lossy by design, e.g. the process of converting

debugging sessions into atomic facts discards causal chains, temporal context, and reasoning paths [13].

3.2 Self-Managing Memory

Self-Managing Memory, such as Letta [11], treats the LLM itself as an operating system managing its own memory via explicit read/write/search operations over a hierarchical store. The agent decides what to remember and how to organize it. Self-managing memory excels at preserving *behavioral continuity*, i.e. reasoning on goals, recurring patterns, and evolving state, corresponding to the *experiential knowledge* and *auto-generated memories* dimensions in Table 1. For software engineering agents that must maintain an evolving picture of project architecture, track long-running refactoring goals, or remember what patterns emerged during debugging, this is the strongest fit, although less transparent than explicit stores: the agent’s internal state cannot easily be inspected, audited, or corrected by developers.

3.3 Graph-Based Temporal Memory

Graph-Based temporal approaches, such as Graphiti [12], can distinguish *event time*, i.e. when something happened, from *ingestion time*, i.e. when the system learned it, with every edge carrying explicit validity intervals. This *bi-temporal* approach is unique among production memory systems and directly addresses the *temporal versioning* and *bi-temporal validity* dimensions that remain near-universally absent in Table 1. Graph-based memory excels at *causal reasoning*. For software engineering tasks requiring understanding of why a decision was made, how events connect across time, or what caused a regression, temporal graphs provide the most natural representation. However, graph construction adds latency and complexity.

3.4 Hybrid and Emerging Approaches

The LoCoMo-Plus results also make a compelling case for hybrid architectures, since no single paradigm masters all competency types [9], and it confirms that the systems capture genuinely different aspects of the memory problem. To this end several emerging directions point toward future software engineering relevant architectures. Reinforcement learning based memory management, such as Memory-R1 [26] and AgeMem [27] replace heuristic memory policies with trained ones. The learned policies discover non-obvious strategies such as preemptive summarization before the context window fills, suggesting a path toward adaptive memory management tuned to software engineering specific reward signals such as issue resolution rate, code quality metrics. Multi-graph memory, as in MAGMA [23], represents each memory item across orthogonal semantic, temporal, causal, and entity graphs with policy-guided traversal, offering fine-grained control relevant for software engineering, e.g. querying dependency relationships,

Property	Extraction	Self-Managing	Graph-Temporal	Hybrid
Factual recall	✓	◇	◇	✓
Behavioral continuity	✗	✓	◇	✓
Causal reasoning	✗	◇	✓	✓
Temporal grounding	✗	✗	✓	◇
Transparency / auditability	✓	✗	✓	◇
Setup complexity	Low	Medium	High	High
Representative system	<i>Mem0</i>	<i>Letta</i>	<i>Graphiti</i>	<i>MAGMA</i>

✓ Supported ◇ Partial ✗ Absent

Table 2. Comparison of memory architecture paradigms across key properties relevant to software engineering agents.

decision provenance, or API validity independently. A formal theory of hierarchical memory [46] and its implementation TiMem [24] define extraction, coarsening, and traversal operators over temporal memory trees, potentially essential for agents processing long execution traces. Hybrid local/cloud architectures are also emerging: LightMem [43] delegates routing and filtering to small local models while HyMem [44] achieves comparable performance at reduced cost via dynamic two-tier retrieval. Since the majority of MCP memory servers support fully local operation and frameworks like Mem0, Letta, and Cognition run on Ollama, sovereign deployments are feasible, though reliable MCP tool calling requires 14B+ parameter models [35,17].

4 Memory Mechanisms for Software Engineering

Beyond the general-purpose memory architectures compared in Table 2, several mechanisms have been developed specifically for or with direct relevance to software engineering agents. These address capabilities that remain largely absent in the production landscape surveyed in Table 1, particularly temporal versioning, decision provenance, and cross-session learning.

4.1 Versioned Context Management and Decision Provenance

The Git Context Controller (GCC) [14] introduces Git-inspired operations for agent memory, addressing the *intra-session* memory problem, i.e. managing working memory within a single task execution. The Lore protocol [15] addresses the complementary *inter-session* problem as it repurposes git commit messages as structured decision records carrying constraints, rejected alternatives, agent

directives, and verification metadata via native git trailers, introducing queryable decision provenance. The practical need for decision provenance is acute. For example, each additional deviation in a mutating action reduces success odds [30], with errors growing as context length increases and agents drift from stale constraints, such as deprecated APIs [28,29].

4.2 Experiential and Cross-Session Learning

The SWE Context Bench [21] evaluates whether agents benefit from prior experience with one critical finding, i.e. summarized experience improves resolution, while unfiltered experience hurts. Several approaches address this through structured learning from trajectories. Experiential Co-Learning [16] mines historical trajectories, Experiential Reflective Learning [17] distills transferable heuristics, and Subtask-Level Memory [19] aligns storage and retrieval at the subtask rather than instance level. MemGrad [20] combines retrospective failure-mode capture with prospective gradient-derived strategies. Procedural memory, such as MemP [37], distills trajectories into step-by-step instructions and higher-level abstractions, with procedural knowledge transferring effectively from stronger to weaker models. ReasoningBank [47] distills generalizable reasoning strategies from self-judged successes and failures, while persistent note-taking [48] records cross-session hindsight notes. Live-SWE-agent [18] self-creates new tools during execution, representing procedural memory through self-modification. Notably, WebCoach [35] reveals a capacity threshold, i.e. 7B models do not benefit from experiential memory, while 32B+ models show pronounced gains, and self-generated experiences outperform externally seeded ones [17].

4.3 Knowledge Graphs for Software Engineering Agent Memory

A comprehensive survey on graph-based agent memory [22] identifies four lifecycle phases, from extraction, that is converting raw data into graph nodes and edges, followed by storage, i.e. organization and indexing, then retrieval, i.e. query-adaptive traversal, and evolution, i.e. update, consolidate, forget. For software engineering agents, graph memory offers three specific advantages. First, they naturally implement *dependency tracking* via relationships for inherently graph-structured code repositories, including files import modules, classes inherit from others, functions call functions [36]. Second, they provide *decision provenance*. Architectural decision records, code review discussions, and commit rationale form a causal graph of project decisions. Graph-based memory can model and preserve the reasoning behind code structure, enabling agents to understand constraints before proposing changes. Third, they lay a foundation for monitoring *temporal evolution*. For example, Graphiti’s bi-temporal model maps naturally onto software evolution, such as an invalid API, a recently adopted coding convention or deprecated dependency. GAM [25] demonstrates the value of hierarchical graph memory showing significant improvements over extraction-based approaches by decoupling episodic from semantic consolidation via an event progression graph.

4.4 Collaborative and Team Memory

Software engineering is inherently collaborative, yet memory research exhibits a strong single-agent bias [6]. Collaborative Memory [31] provides multi-user memory sharing with private and shared tiers, where each fragment carries immutable provenance attributes. This maps naturally onto software engineering teams, i.e. private developer context versus shared project knowledge. However, MINJA [32] demonstrates that shared-memory agents are vulnerable to injection attacks, a critical supply-chain risk for agents with codebase access. SSGM [33] and MemArchitect [34] propose governance middleware, while the mnemonic sovereignty survey [49] addresses EU AI Act compliance. Yet no approach targets software engineering-specific governance needs such as repository-aligned access control or temporal policies tied to release cycles.

5 Discussion

Synthesizing our analysis, we identified several open gaps specific to memory for software engineering agents as inspirations for future research. Software engineering agents discard resolution trajectories after each session. Early research solutions exist, but no production harness or community add-on integrates cross-session learning. No production software engineering agent and only a tiny fraction of community add-ons track temporal knowledge validity. No software engineering specific temporal memory exists. Also, there exists only one software engineering specific temporal benchmark. Convention retention, experience transfer, staleness detection, and decision provenance remain unevaluated. Collaborative memory is rarely addressed, yet software engineering teams need shared decisions, private debugging notes, and repository-aligned access control. Research on memory injection attacks demonstrates that shared-memory agents lack adequate governance and security, while GDPR-compliant deletion remains entirely unaddressed. Software engineering craft knowledge is overwhelmingly procedural, yet procedural memory is the rarest type in both research and the community ecosystem. The MCP ecosystem provides a composable memory architecture where each server addresses a different CoALA memory type. Yet no orchestration layer decides which memory to consult when, resolves conflicts between overlapping stores, or manages lifecycle across layers. Assembling the full stack remains a manual integration challenge.

Our analysis has limitations as it represents a snapshot as of mid 2026 in a rapidly evolving landscape, evaluations in Table 1 are based on documentation and hands-on testing rather than controlled experiments, and our selection of MCP add-ons targets representativeness rather than completeness. Nevertheless, the convergent evidence across production harnesses, community add-ons, and research literature consistently points to the same structural deficits. Closing these gaps will require a coordinated effort and a clear vision of transforming software engineering agents from amnesic tools into learning collaborators that grow alongside the software they help build.

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